

Environment-Adaptive Bluetooth Speaker

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ECE Makerspace Student Lab Assistant Design Project

1. Design Concept

For this project, I designed a compact Bluetooth speaker built around a truncated square-pyramid enclosure approximately 6×6 inches at its base and 4.5 inches tall.

I initially chose the pyramid shape because it just seemed convenient since it meant that assembly could most likely be easier, looks visually appealing, and would have the potential to potentially increase audio quality since speakers are usually in my experience placed somewhere below ear level so having audio be emitted at a slight angle could help counter this, also something I never really liked was thinking that the surrounding surface the speaker could be placed on could disrupt audio quality due to sound bouncing off the surface. As I developed the design, however, I realized that the four angled sides could also serve a functional purpose.

Instead of having all of the sound originate from one face, each side can act as its own directional speaker surface.

My current design uses four small full-range speaker drivers, one on each angled face, allowing sound to be projected in four directions. A compact distance sensor associated with each face would allow the speaker to determine whether that side is facing open space or is positioned close to an obstruction such as a wall.

When the adaptive mode is enabled, a microcontroller would use these measurements to reduce the output of a driver facing a nearby obstruction. For example, if Side 3 were placed directly against a wall, the system could reduce or disable Driver 3 while allowing the other three drivers to continue operating.

The goal is to avoid sending unnecessary audio power directly toward an obstruction and potentially reduce unwanted reflections from very nearby surfaces. The actual acoustic and power benefits would need to be measured and refined through physical prototyping.

The speaker would also include Bluetooth connectivity, rechargeable power through USB-C, physical controls, and a normal operating mode in which environmental adaptation can be disabled.

2. Physical Design

Enclosure

The enclosure is a truncated square pyramid with:

- Base: approximately 6×6 in
- Height: approximately 4.5 in
- Top dimensions: 4×4 in

As far as I know, there are no free, widely available materials I would need for this project for the speaker, for the best material possible, so some alternatives would be to just look for anything available within the scrap bin in the design and innovation building or to purchase

online directly unless there is some MDF, acrylic, and/or plywood available. If only the worst options are available, stuffing may be necessary to control internal echo and smooth out the audio.

- Primary enclosure material: tbd; ideally plywood
- Approximate wall thickness: ½ in

The wider bottom provides the speaker with a stable footprint and space for the battery and other heavier components. The four angled faces provide mounting surfaces for the four speaker drivers.

The smaller flat surface at the top becomes the control panel.

Internal Structure

Inside the enclosure, I designed four structural posts positioned near the four corners.

These posts connect the wider bottom structure to the smaller top structure and provide a rigid internal skeleton for the enclosure. They could also provide mounting locations for the exterior panels and top control panel.

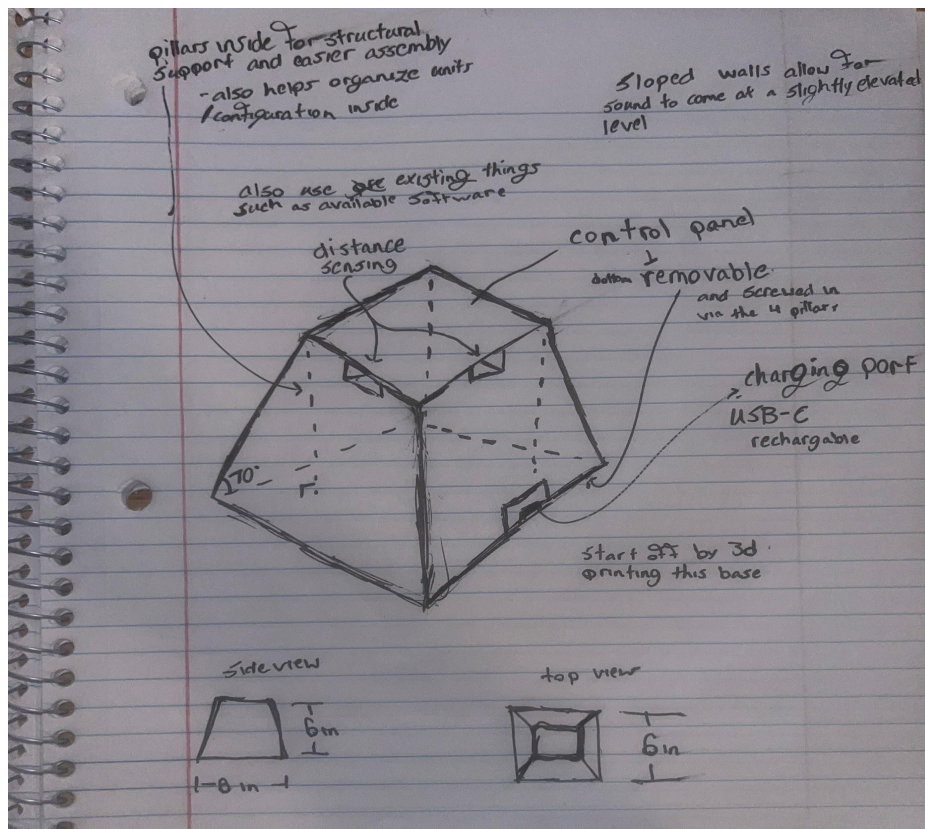
The bottom of my current CAD model contains screw holes so that once assembly is complete, it can be screwed in and is available for check up/maintenance whenever the speaker has any issues. I chose to place these fasteners toward the bottom because it makes assembly and fixing easier for fingers to fix the issue instead of having to dismantle the whole thing and reglue wood panels on.

Control Panel: The flat top surface acts as the main user interface.

My proposed controls include:

- Power
- Volume
- Bluetooth pairing/status
- Adaptive sensing mode ON/OFF
- Battery/status indicator
- Skip, pause, or go back

The bottom could also be designed as a removable service panel, allowing access to the internal electronics without disassembling the entire enclosure



- Rough idea

Figure 1. Overall truncated-pyramid enclosure and structural design.

3. Audio System

The speaker receives audio wirelessly from a phone, computer, or other Bluetooth-enabled device.

The basic signal path is:

Bluetooth Device → Bluetooth Audio Receiver → Audio Processing/Control → Amplification
→ Speaker Drivers

Four small full-range drivers are preferred, with one driver mounted on each angled face.

Driver size: ~2 in with a diameter less than or equal to 2.5 in

Driver model/type: Dayton Audio CE Series CE53N-4 2" Dual Neo Full-Range Speaker Driver - probably have to order it online myself unless the makerspace has anything available (quality of driver is not my main focus since I'm just trying to optimize for the worst possible one)

Number of drivers: 4

Using four independently controllable outputs is important because the environmental sensing feature requires the system to alter one direction without necessarily affecting the other three.

Rather than designing the Bluetooth receiver and amplifier completely from scratch, I would use existing audio modules where appropriate and focus on integrating them with the enclosure, power system, sensors, and control software.

Sketch

Example labels:

- Driver 1 — Front
- Driver 2 — Right
- Driver 3 — Rear

- Driver 4 — Left
-

4. Environment-Adaptive System

The experimental feature of the speaker is its ability to respond to its physical surroundings.

Distance Sensing

My original idea was to use ultrasonic/sonar sensors. However, conventional ultrasonic modules would occupy significant space on an enclosure this small.

Instead, I would investigate using compact time-of-flight (ToF) distance sensors. These sensors measure distance using infrared light and could be integrated much more discreetly into the four faces.

Each sensor would measure the available space in approximately the same direction as its corresponding driver.

For example:

Sensor 1 → Driver 1

Sensor 2 → Driver 2

Sensor 3 → Driver 3

Sensor 4 → Driver 4

Control Logic

The sensors communicate their measurements to a microcontroller.

The software would classify each face as either open or obstructed based on a distance threshold determined experimentally.

For example:

Open space → normal driver output

Nearby obstruction → reduced driver output

If Driver 3 were facing a nearby wall:

Sensor 3 detects wall → Microcontroller identifies Side 3 as obstructed → Audio control reduces Channel 3 → Drivers 1, 2, and 4 continue operating

Initially, I would implement this as a relatively simple threshold system. A more advanced version could gradually change output based on measured distance instead of simply switching between two states.

Why Include an Adaptive Mode?

The goal is to determine whether changing the output direction based on speaker placement can:

1. Reduce unnecessary power consumption from drivers pointed directly toward nearby obstructions.
2. Reduce unnecessary sound being projected directly into nearby surfaces.
3. Allow the speaker to automatically respond when its position within a room changes.

These would be design hypotheses rather than assumed benefits. If I physically constructed the speaker, I would test different distances, output levels, sensor positions, and speaker placements

to determine whether the system actually produces meaningful improvements.

The user could disable this feature through the top control panel, returning the speaker to normal operation with all four drivers active.

5. Electronics and Software Architecture

The speaker contains three major electronic systems:

Audio

Bluetooth Receiver → Audio Processing → Amplifier(s) → Four Drivers

The audio system needs independently controllable channels so that individual drivers can be reduced when necessary.

Proposed audio module/amplifier: 2.0-channel class D Bluetooth amplifier

Environmental Control

4× ToF Sensors → Microcontroller → Audio Channel Control

Proposed microcontroller: ESP32

Proposed ToF sensor: ToF sensor VL53L0X

The microcontroller would periodically read the four distance sensors and determine whether each face is obstructed.

Power

The speaker would contain a rechargeable battery and USB-C charging.

The general power system would be:

USB-C → Charging/Battery Management → Rechargeable Battery → Speaker Electronics

The battery system supplies power to:

- amplifier/audio electronics;
- Bluetooth system;
- microcontroller;
- distance sensors;
- indicators/controls.

Battery: Rechargeable lithium-ion battery pack using 18650 cells. The final series/parallel configuration and capacity would be selected after determining the operating-voltage and current requirements of the amplifier system.

Charging/power module: USB-C charging and battery-management system (BMS) matched to the selected lithium-ion battery configuration, providing charging, overcharge, over-discharge, and overcurrent protection.

I would position heavier components such as the battery toward the wide bottom of the enclosure to keep the center of gravity relatively low.

Ideal diagram:

Phone → Bluetooth → Audio Processing → Amplification → D1 / D2 / D3 / D4

and

ToF 1 / 2 / 3 / 4 → Microcontroller → Audio Control

(D = driver)

6. Components

Component	Quantity	Purpose
Full-range speaker drivers	4	Produce audio in four directions
Bluetooth audio receiver/module	1	Receives wireless audio
Audio amplifier/module(s)	1	Provides sufficient power to the drivers
Microcontroller	1	Processes sensor measurements and controls adaptive behavior

ToF distance sensors	4	Detect nearby obstructions around each face
Rechargeable battery	1	Portable power
USB-C charging/BMS hardware	1	Battery charging and protection
Volume control	1	User volume adjustment
Buttons/switches	7	Power, Bluetooth, adaptive-mode controls
Status indicator(s)	3	Bluetooth/battery/mode feedback
Enclosure panels	6	Main speaker enclosure
Structural posts	4	Reinforce and connect enclosure structure
Speaker grilles	4	Protect drivers

Wiring/connectors	As needed	Electrical connections
Screws/fasteners	As needed	Assembly and serviceability
Gasket/damping material	As needed	Reduce vibration/rattling around mounted components

7. Construction Using the ECE Makerspace

If I were to physically construct the speaker, I would begin by finalizing the enclosure in CAD and checking that the drivers, electronics, battery, structural posts, sensors, and wiring fit inside the available volume.

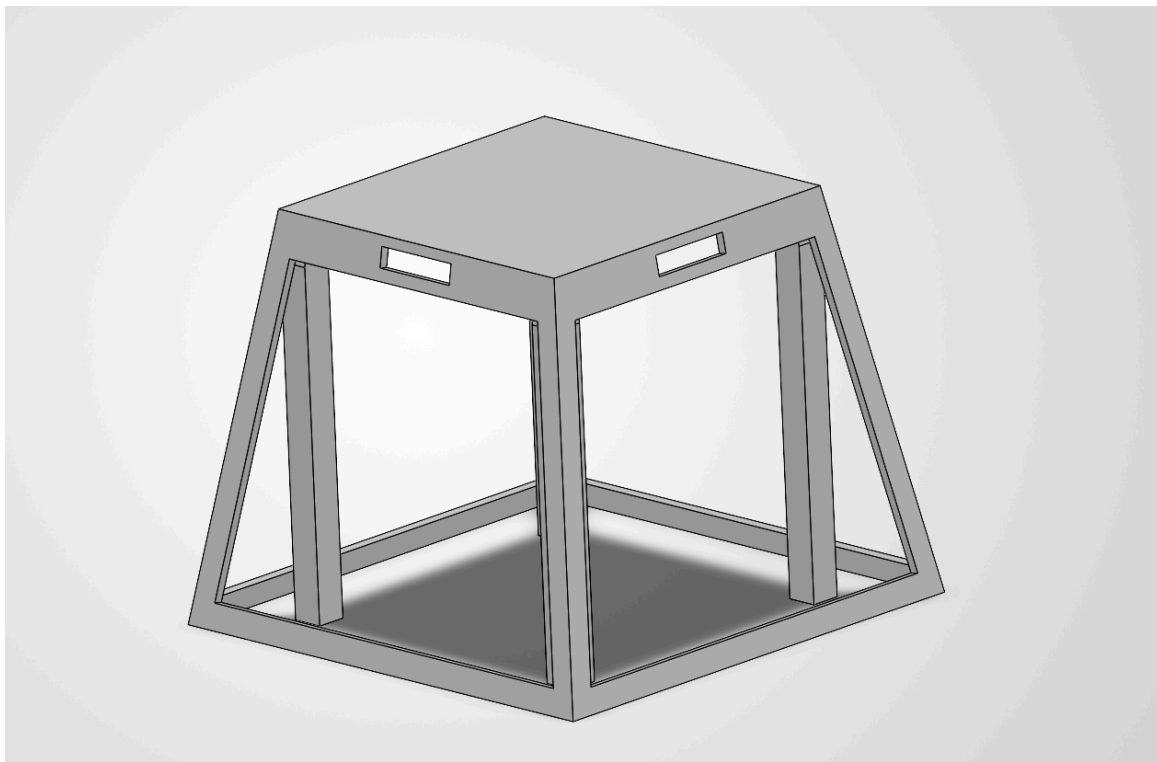
Step 1 - Finalize the CAD Design

I would use Autodesk Inventor to finalize:

- enclosure dimensions;

- angled exterior panels;
- driver openings;
- structural-post locations;
- screw holes;
- control-panel dimensions;
- sensor mounting locations;
- electronics mounting points.

I would also verify that the internal components do not interfere with one another.



- Rough idea

Step 2 - Fabricate the Enclosure

I would use the Makerspace's CNC router and/or woodworking equipment to cut the primary enclosure pieces from the wood needed to enclose the driver and frame the 3d print.

The CNC router would be particularly useful for accurately creating the driver openings and other geometry determined from the CAD model. I could use the laser cutter for thinner pieces such as speaker grilles, control-panel components, templates, or decorative elements where appropriate.

Step 3 - Produce Custom Mounts

I would use 3D printing for smaller custom components that would be inconvenient to manufacture from the main enclosure material, potentially including:

- ToF sensor mounts;
- circuit-board brackets;
- control/button mounts;
- cable-management pieces;

Step 4 - Assemble the Structure

The enclosure panels would be attached around the four-post internal frame.

The lower structure and existing screw locations would be used. Before permanently closing the enclosure, I would check panel alignment, driver clearance, and accessibility to the electronics.

Step 5 - Install Electronics

Using the Makerspace's soldering equipment, I would connect the:

- speaker drivers;
- amplifier;
- Bluetooth audio hardware;
- microcontroller;
- distance sensors;
- battery/charging system;
- control-panel components.

Wiring would be secured internally to prevent loose wires from contacting the drivers or vibrating against the enclosure.

Step 6 - Program and Test

I would first test the speaker as a conventional Bluetooth speaker before enabling the adaptive system.

After confirming normal audio operation, I would program the microcontroller to read each distance sensor and adjust the appropriate audio channel.

I would then test:

- different obstruction distances;
- wall versus corner placement;
- sensor thresholds;

- different levels of channel reduction;
- power consumption;
- audible effects of the adaptive system;
- vibration/rattling;
- enclosure stability.
- Any leakage

The results of these tests would determine whether the original adaptive behavior should be modified.

8. Design Challenges and Future Development

If I continued developing the design, one of the first areas I would investigate would be the acoustic enclosure itself.

Four drivers sharing a small internal volume could interact with one another, particularly when some drivers are active while others have reduced output. I would investigate whether internal partitions or individual acoustic chambers are necessary.

I would also experiment with:

- driver size and placement;
- internal enclosure volume;
- acoustic damping;
- different obstruction thresholds;
- gradual rather than binary channel adjustment;

- power consumption with adaptive mode enabled versus disabled;

A more advanced version could also use microphones to directly measure the acoustic response of the room. The current ToF system is intentionally simpler: it measures the speaker's physical surroundings rather than claiming to directly measure room acoustics.

9. Final Design Summary

My design combines a relatively simple Bluetooth speaker with an experimental environmental-control system.

The truncated-pyramid geometry provides four directional surfaces for audio output, a wide and stable lower structure, and a flat top for user controls. Internally, four structural posts reinforce the enclosure and provide mounting support.

Four distance sensors allow the speaker to estimate whether each output direction is open or obstructed. A microcontroller uses that information to control the corresponding audio channels, allowing the speaker to reduce output toward nearby obstructions while continuing to produce sound in open directions.

The project combines CAD, fabrication, electronics, embedded programming, audio, and iterative testing rather than relying on any single discipline.

If I physically built the design, I would expect the first version to reveal problems I have not anticipated. I would use those results to iterate on the enclosure, sensor thresholds, audio control, and internal layout rather than treating the initial design as a finished solution.

